

COLES COUNTY AP, IL, USA

WMO: 725317

Lat: 39.478N

Lon: 88.280W

Elev: 721

StdP: 14.32

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 53802

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.8	7.4	-5.2	4.3	4.1	-0.4	5.5	8.4	29.0	23.8	26.6	24.4	9.1	290	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	91.4	75.3	89.5	75.2	87.2	74.1	78.9	87.6	77.5	85.7	76.1	83.9	7.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.4	141.6	84.4	74.9	134.6	82.7	73.4	127.4	80.9	43.0	87.8	41.4	85.7	40.0	83.7	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.5	20.8	18.7	-4.5	95.6	6.1	4.8	-8.8	99.0	-12.4	101.8	-15.8	104.5	-20.2	107.9		
(5)			-5.1	81.6	5.8	1.2	-9.3	82.4	-12.7	83.1	-16.0	83.8	-20.2	84.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.8	28.0	31.9	42.6	54.5	64.5	72.9	75.5	74.1	68.0	56.0	43.8	32.9		
	DBStd	19.12	12.00	11.63	11.21	9.37	8.33	5.89	5.66	5.12	7.31	9.30	10.06	11.02		
	HDD50	2370	685	515	287	60	4	0	0	0	44	237	538			
	HDD65	5222	1147	927	696	332	116	10	1	3	51	308	637	994		
	CDD50	3773	3	8	59	193	454	687	792	748	540	229	51	9		
	CDD65	1148	0	0	3	16	100	247	328	285	140	28	1	0		
	CDH74	10079	0	0	13	115	760	2112	3137	2474	1249	215	4	0		
	CDH80	3396	0	0	0	9	170	705	1188	855	422	47	0	0		
(14) Wind	WSAvg	8.8	10.6	10.7	10.6	10.6	8.6	7.2	6.2	5.9	6.6	8.6	9.7	10.0		
(15) Precipitation	PrecAvg	41.7	2.4	2.2	3.4	4.0	4.5	3.9	4.5	3.3	3.3	3.3	3.8	3.1		
	PrecMax	56.6	7.7	5.6	7.7	8.6	11.6	10.4	10.4	10.7	8.9	9.3	11.9	8.8		
	PrecMin	28.1	0.0	0.2	1.0	0.2	0.8	0.7	1.3	0.6	0.1	0.3	0.6	0.4		
	PrecStd	6.7	1.6	1.3	1.8	2.1	2.4	2.2	2.6	2.1	2.4	2.0	2.2	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.2	65.4	75.9	81.4	87.8	92.6	98.0	93.7	92.1	84.9	73.0	63.8		
		MCWB	57.4	58.8	63.0	65.6	72.1	73.5	76.7	76.4	71.7	70.5	60.5	59.4		
	2%	DB	54.9	59.3	70.2	77.4	84.2	89.6	92.1	90.4	88.5	80.2	67.4	58.4		
		MCWB	51.7	54.6	60.0	63.5	70.0	74.3	77.2	75.8	72.1	66.6	59.4	55.1		
	5%	DB	50.2	54.3	65.0	73.5	81.6	87.2	89.5	87.8	85.1	75.2	63.3	53.5		
		MCWB	46.8	49.3	56.7	61.8	68.3	73.5	76.9	74.4	70.9	63.6	56.5	50.4		
	10%	DB	44.9	48.6	60.9	70.4	78.7	84.1	86.5	84.5	81.4	71.6	59.2	48.0		
		MCWB	41.3	44.1	54.0	60.0	66.9	71.8	75.5	72.8	68.1	62.7	53.1	44.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.8	59.9	64.4	69.1	74.7	79.0	81.3	80.3	76.4	72.3	64.3	60.6		
		MCDB	60.6	64.8	73.7	77.6	83.8	87.6	90.3	88.8	86.1	82.1	69.3	63.8		
	2%	WB	52.8	55.3	61.6	66.3	72.5	77.0	79.7	78.2	74.6	69.8	60.6	55.4		
		MCDB	54.6	58.2	68.6	73.9	81.2	85.3	88.7	86.7	83.7	76.1	65.0	57.9		
	5%	WB	47.1	50.1	58.5	63.9	70.6	75.4	78.2	76.5	72.8	67.0	57.8	50.4		
		MCDB	49.7	54.1	64.2	70.8	78.7	83.6	86.5	84.5	80.7	72.6	62.9	52.8		
	10%	WB	41.1	44.3	54.6	61.5	68.8	74.0	76.9	75.0	71.0	63.5	53.6	45.2		
		MCDB	44.9	48.4	60.1	68.5	76.0	81.7	84.6	82.2	78.1	69.2	58.6	47.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	15.6	17.8	19.6	18.5	18.2	18.2	19.1	21.4	19.3	16.4	13.3		
		MCDBR	20.2	21.2	21.9	22.3	20.6	20.5	19.9	21.1	23.4	23.0	20.0	16.8		
		MCWBR	17.6	17.2	14.2	13.0	10.2	9.3	8.5	9.4	9.5	12.0	14.0	14.5		
	5% WB	MCDBR	19.6	20.4	19.6	19.8	18.4	17.5	17.5	17.9	18.9	18.9	18.6	16.0		
		MCWBR	18.2	17.9	13.7	13.4	10.1	9.3	9.0	9.5	8.9	11.3	14.9	14.9		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.320	0.353	0.398	0.435	0.436	0.460	0.460	0.416	0.361	0.315	0.301		
	taud		2.449	2.388	2.376	2.249	2.196	2.231	2.154	2.140	2.251	2.404	2.490	2.474		
	Ebn at Noon		276	285	286	279	269	268	260	257	261	266	269	268		
	Edh at Noon		25	31	35	43	46	44	47	47	39	30	24	23		
(44) All-Sky Solar Radiation	RadAvg		572	839	1161	1520	1730	1997	1945	1752	1472	1035	690	485		
	RadStd		66	90	98	95	153	128	121	113	103	107	72	58		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+178	+126

Nomenclature: See separate page